

SAP on RHEL

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Home

Introduction

Course Title: SAP on RHEL

Description: Course description FIXME

Duration: FIXME hours

Topics covered in this course

- Overview of RHEL for SAP Solutions
- Red Hat Portfolio for SAP
- SAP HANA Infrastructure

Prerequisites

This course assumes that you have the following prior experience:

- Course or experience...
- Course or experience...
- Course or experience...



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Lab Environment

FIXME: Instructions for accessing the lab environment for this hands-on based training.

Overview of RHEL for SAP Solutions

Red Hat Enterprise Linux (RHEL) for SAP Solutions provides a standardized, automated infrastructure designed to support SAP workloads, including SAP S/4HANA, across on-premises and cloud environments.

Key components of the RHEL for SAP Solutions subscription include:

- **RHEL Foundation:** The core enterprise Linux operating system.
- **High Availability:** The High Availability Add-on for maintaining continuous SAP service uptime.
- **Management & Insights:** Tools for lifecycle management (Smart Management), proactive risk analytics, and automated mitigation (Red Hat Insights).
- **Automation:** RHEL System Roles for SAP and integration with Red Hat Ansible Automation Platform to streamline SAP deployments and day-2 operations.
- **SAP Optimization:** Pre-configured SAP-specific libraries and configurations to ensure optimal system performance.
- **Extended Lifecycle:** Update Services for SAP Solutions, providing support for specific RHEL minor releases for up to four years.

Introduction to Red Hat Enterprise Linux for SAP

Red Hat Enterprise Linux (RHEL) for SAP Solutions is a purpose-built operating system environment designed to meet the rigorous performance, stability, and reliability requirements of SAP workloads. As organizations migrate to SAP S/4HANA or modernize their existing Digital Core, RHEL serves as the foundation for an Open Hybrid Cloud strategy, allowing for consistent operations across on-premises data centers and public cloud providers.

Why RHEL for SAP?

SAP applications are mission-critical, often serving as the central nervous system of an enterprise. Deploying these workloads on standard Linux distributions can lead to performance bottlenecks or lack of specialized support. RHEL for SAP Solutions addresses these challenges by bundling critical components specifically tuned for the SAP ecosystem.

Unlike a generic OS, the RHEL for SAP Solutions subscription provides an automated, end-to-end infrastructure specifically engineered to minimize downtime, simplify management, and ensure compliance with SAP's technical requirements.

Key Components of the RHEL for SAP Portfolio

The RHEL for SAP Solutions subscription is more than just an OS; it is a comprehensive suite of tools designed to manage the entire lifecycle of an SAP environment:

- **Red Hat Enterprise Linux:** The core, stable foundation optimized for high-performance enterprise applications.

- **High Availability (HA) Add-on:** Provides the clustering framework necessary to ensure SAP services (like SAP HANA or SAP NetWeaver) remain operational during hardware or service failures.
- **Red Hat Smart Management:** Simplifies lifecycle management, allowing administrators to manage patch levels and configurations across large-scale SAP environments.
- **Red Hat Insights:** An analytics-driven tool that provides proactive optimization, identifies security risks, and suggests automated mitigations before they impact production performance.
- **RHEL System Roles for SAP:** Pre-defined automation roles that enforce consistent configurations (such as kernel parameters and storage tuning) to match SAP best practices during deployment.
- **SAP-specific Software Components:** Optimized libraries and configurations tailored for SAP workload requirements.
- **Update Services for SAP Solutions:** Offers extended support for specific minor RHEL releases—four years from General Availability—enabling SAP customers to maintain stable operating environments while managing longer update cycles.

Automation with Ansible

Beyond the OS components, Red Hat facilitates the full automation of SAP S/4HANA migrations and Day-2 operations through the integration with the **Red Hat Ansible Automation Platform**. By combining RHEL for SAP Solutions with Ansible, IT teams can transform manual, error-prone deployment processes into repeatable, standardized workflows, significantly reducing the "time-to-value" for new SAP initiatives.



For organizations scaling their SAP footprint, leveraging **Red Hat Insights** is crucial. It acts as a continuous health-check tool, mapping your RHEL configuration directly against known SAP support notes and security advisories to prevent potential outages.

Hands-on Activity: Identifying SAP-Specific Components

In this activity, you will verify if the necessary SAP-specific repositories are enabled on your RHEL system.

1. Prerequisites

- A system registered with a valid RHEL for SAP Solutions subscription.
- Root or **sudo** access.

2. **Step 1: Check enabled repositories** Run the following command to view your currently enabled repositories and ensure the SAP-specific repos are active:

```
subscription-manager repos --list-enabled | grep -i sap
```

1. **Step 2: Verify SAP-tuned profiles** RHEL for SAP includes pre-configured **tuned** profiles. Check which profiles are available for your environment:

```
tuned-adm list
```

Look for profiles prefixed with **sap-** (e.g., **sap-hana** or **sap-netweaver**). These profiles automatically adjust kernel parameters, disk I/O schedulers, and memory settings to align with SAP's performance requirements.

1. **Step 3: Apply an SAP profile** If your server is hosting an SAP HANA database, apply the SAP HANA tuned profile:

```
sudo tuned-adm profile sap-hana
```

By executing these steps, you have ensured that your OS is not just running, but is specifically optimized for the SAP HANA workload requirements.

Understanding the RHEL for SAP Solutions subscription

The RHEL for SAP Solutions subscription is more than just an operating system license; it is an integrated, specialized platform designed to meet the rigorous performance, reliability, and lifecycle requirements of SAP environments. By consolidating essential components into a single subscription, Red Hat enables organizations to build a standardized, automated, and resilient foundation for their SAP Digital Core.

Subscription Components

The RHEL for SAP Solutions subscription provides a comprehensive bundle of software and services designed to streamline the deployment and management of SAP workloads. Key components include:

- **Red Hat Enterprise Linux (RHEL):** The core, hardened Linux distribution that serves as the foundation for SAP NetWeaver and SAP HANA workloads.
- **High Availability (HA) Add-on:** A robust clustering framework specifically engineered to support SAP-certified high-availability configurations, ensuring minimal downtime for critical SAP services.
- **Red Hat Smart Management:** Provides integrated lifecycle management, allowing administrators to manage software patches and security updates across large-scale RHEL deployments.
- **Red Hat Insights:** A predictive analytics tool that continuously monitors system health, offering proactive risk assessment and automated mitigation strategies to identify potential issues before they impact business operations.
- **RHEL System Roles for SAP:** A collection of Ansible-based roles that automate the complex configuration tasks required to prepare a RHEL environment for SAP applications, adhering to SAP-specific best practices.

- **SAP-Specific Software Components:** Includes pre-configured libraries and kernel tuning parameters required to meet the SAP "Tuning Guide" requirements for optimal performance.
- **Update Services for SAP Solutions (E4S):** A critical feature for SAP environments that provides long-term support for specific minor RHEL releases (for four years from General Availability), allowing for more flexible maintenance windows and reduced update frequency.

Value Proposition: The Open Hybrid Cloud

By leveraging this subscription, organizations can execute an Open Hybrid Cloud strategy. Because the subscription is consistent across on-premises, private, and public cloud environments, SAP customers can:

1. **Standardize infrastructure:** Use the same RHEL foundation regardless of the deployment site.
2. **Automate operations:** Integrate with Red Hat Ansible Automation Platform to orchestrate SAP S/4HANA migrations and day-2 operations.
3. **Ensure Compliance:** Utilize the included management tools to maintain consistency across the entire SAP landscape.

Hands-on Activity: Verifying Subscription Benefits

In this exercise, you will confirm your system is correctly registered and configured to access the SAP-specific repositories provided by your subscription.



Ensure your system is registered with Red Hat Subscription Management (RHSM) before beginning this activity.

1. **Check Registration Status** Execute the following command to verify your system is registered and the subscription is attached: [source,bash] `--- subscription-manager status ---`
2. **Enable SAP-Specific Repositories** The subscription provides specialized repositories. Verify and enable the required repositories for your RHEL version: [source,bash] `--- # List available repositories to find the SAP-specific ones subscription-manager repos --list | grep -i "sap"`

```
# Enable the RHEL for SAP Solutions repository (example for RHEL 8)
subscription-manager repos --enable=rhel-8-for-x86_64-sap-solutions-rpms
----
```

3. **Verify Insights Connectivity** To leverage the proactive optimization features of your subscription, ensure the Red Hat Insights client is active: [source,bash] `--- insights-client --register ---`
4. **Validate Automation Readiness** Confirm that the `rhel-system-roles` package is installed to begin automating your SAP configurations: [source,bash] `--- dnf install rhel-system-roles ---`

By completing these steps, you have ensured that your environment is properly configured to leverage the support and automation capabilities unique to the RHEL for SAP Solutions subscription.